SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Name of Student : Vaishnavi Pruthviraj More

Roll No : B - 72

Batch : SB4

Course Name and Code: Data Structures Lab (ITL302)

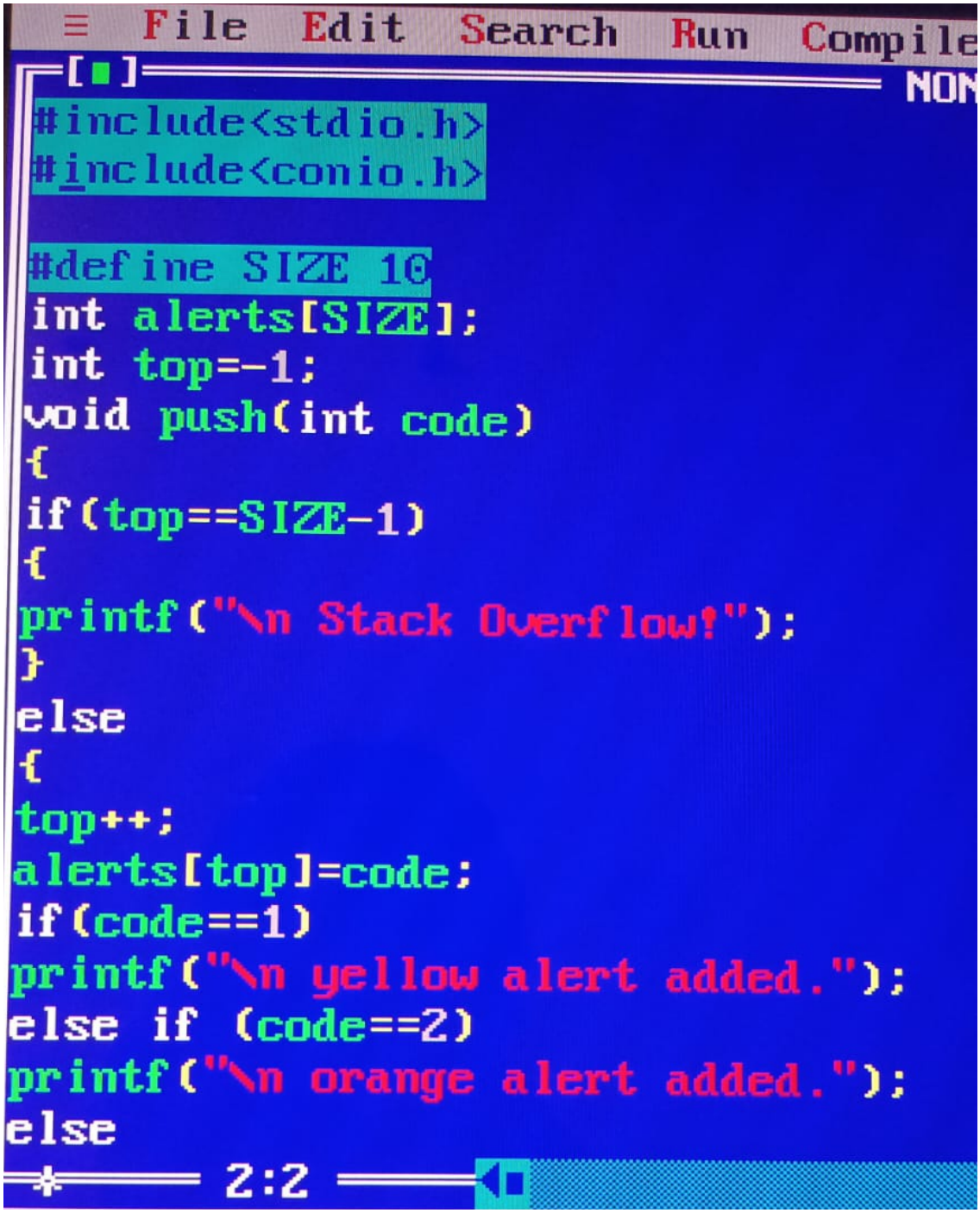
Semester: III (SYIT)

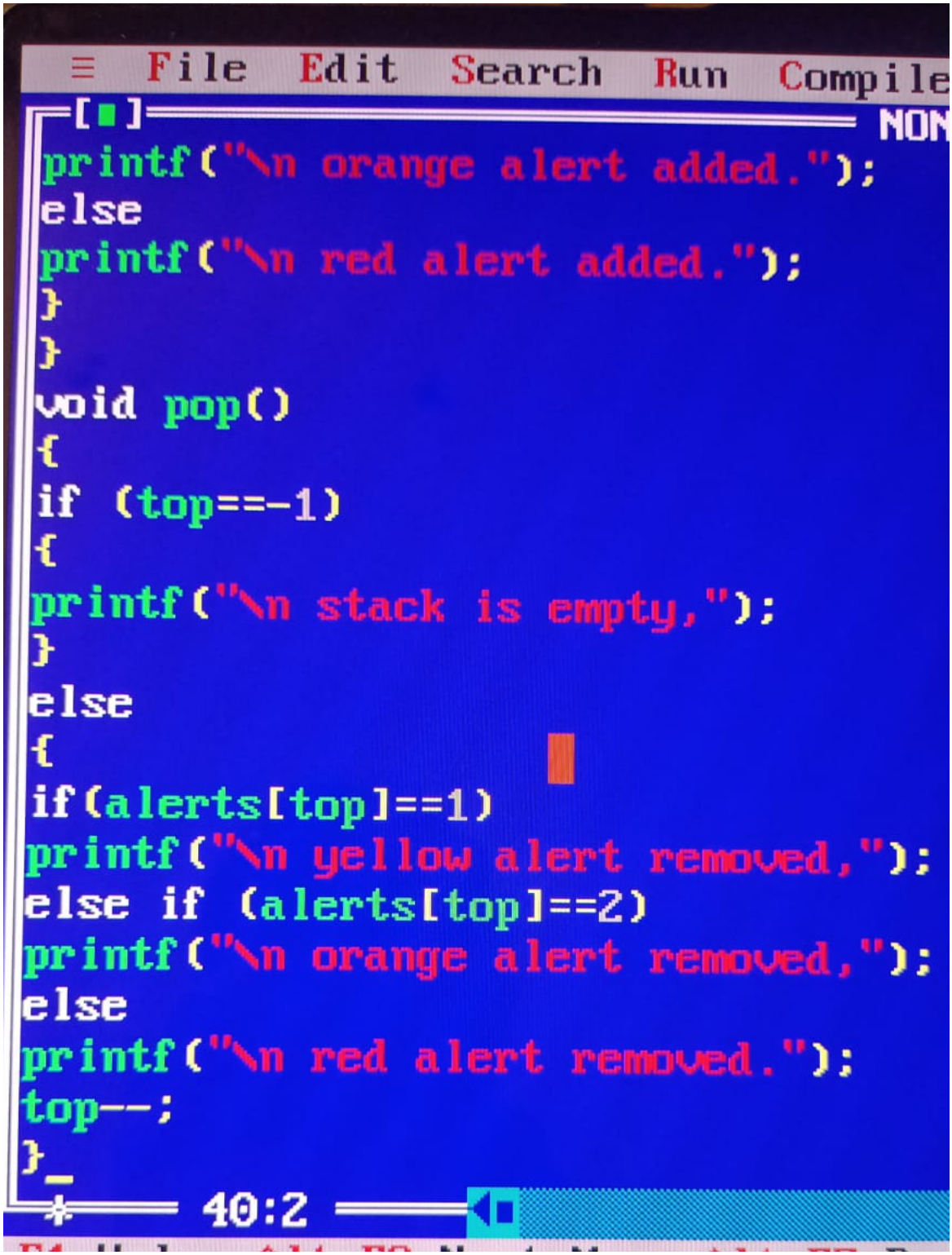
Academic Year: 2026-27 (Odd Semester)

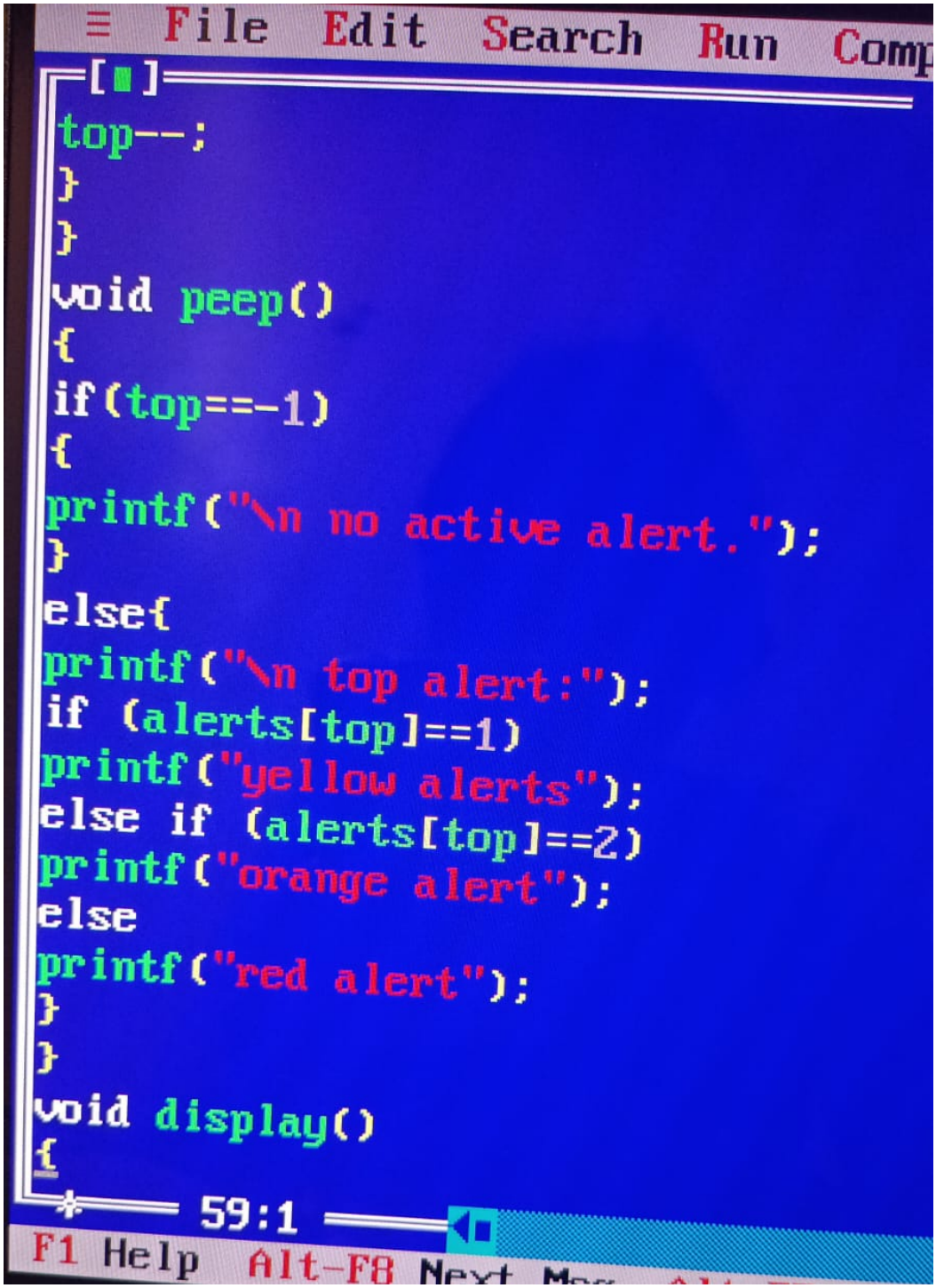
EXPERIMENT 1

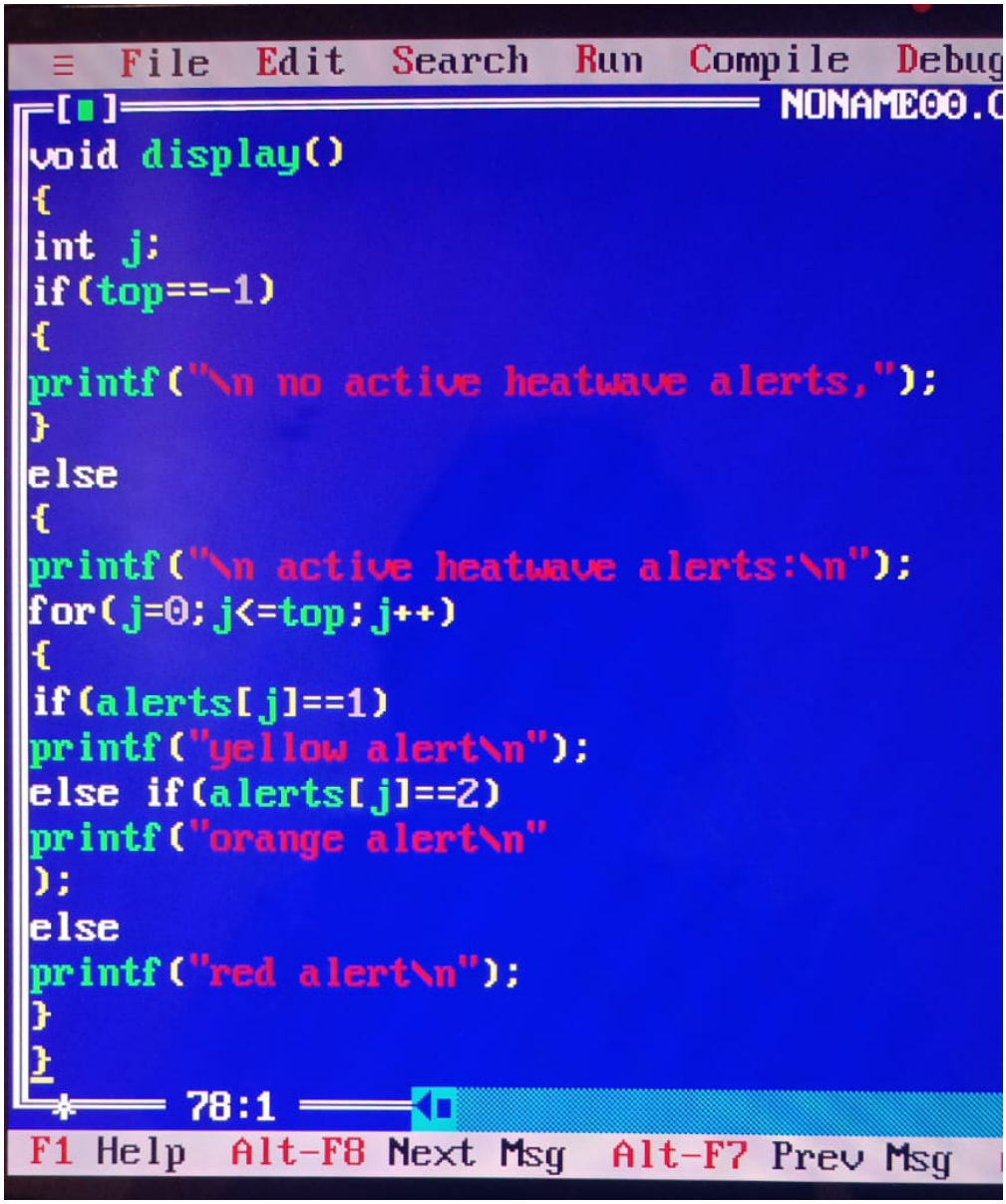
Aim: To implement a stack using an array and apply it to a heatwave alert system that generates and removes alerts based on the given temperature thresholds.

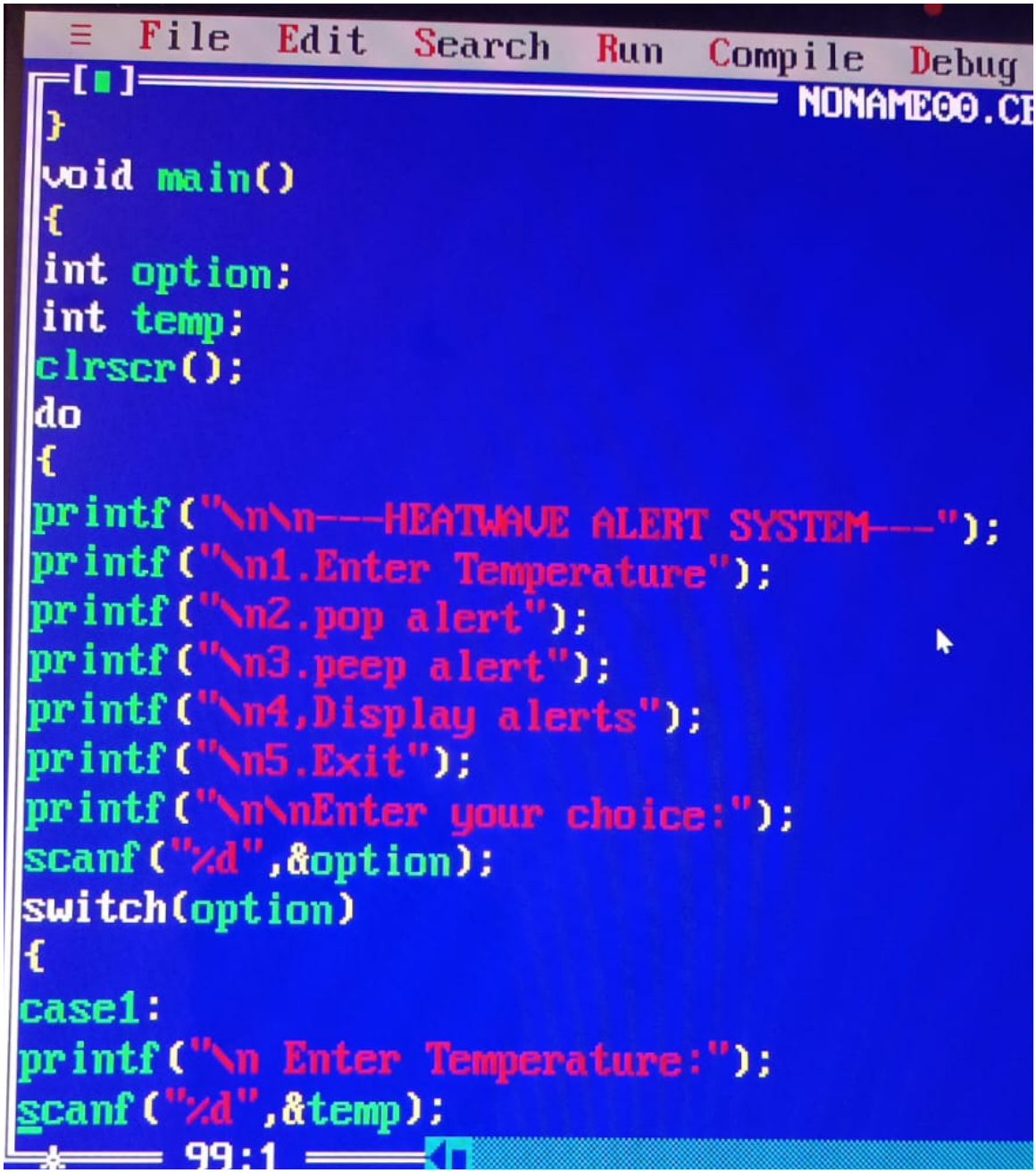
Code:

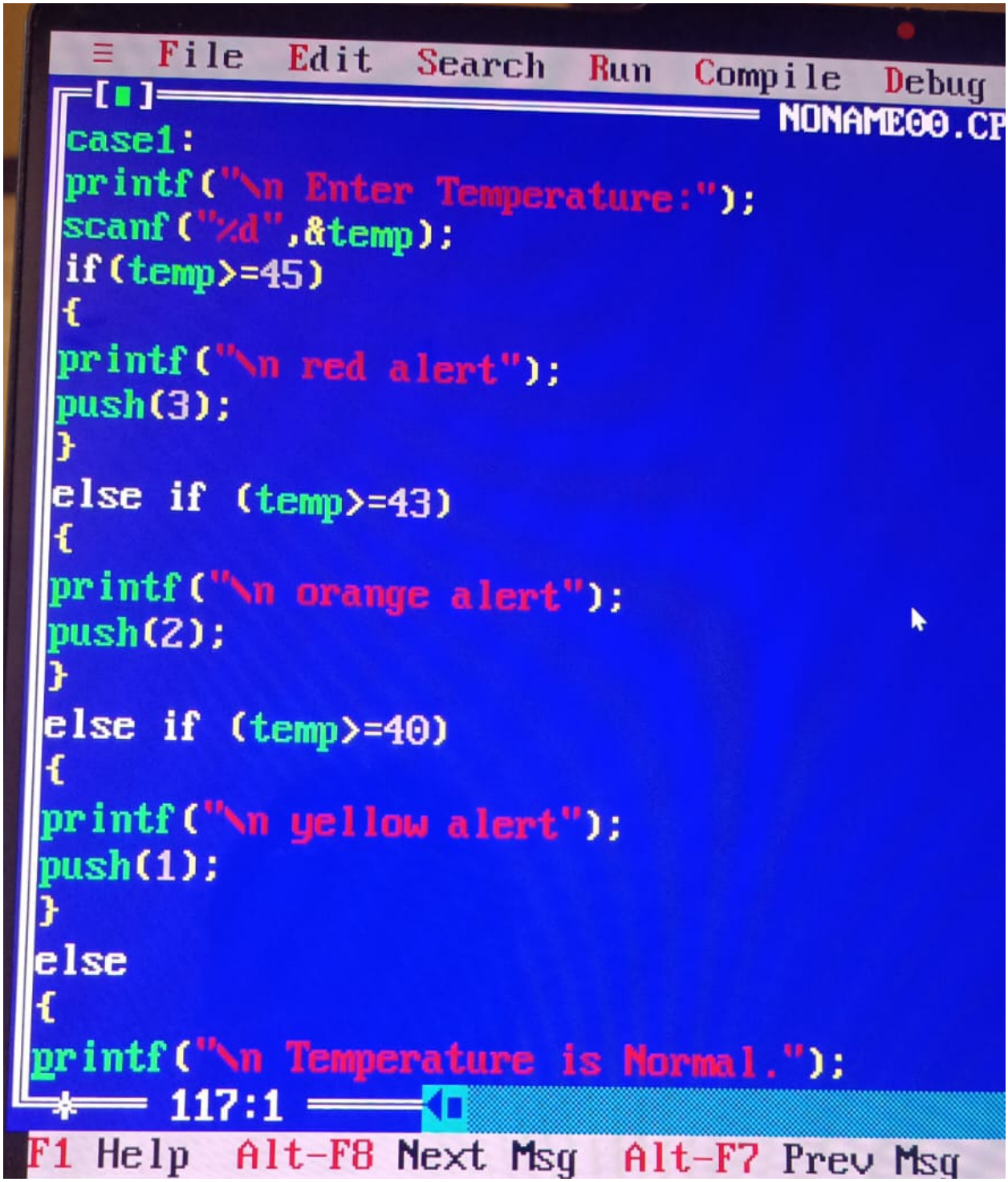

File Edit Search Run Compile
NON
#include<stdio.h>
#include<conio.h>
#define SIZE 10
int alerts[SIZE];
int top=-1;
void push(int code)
{
if(top==SIZE-1)
{
printf("\n Stack Overflow!");
}
else
{
top++;
alerts[top]=code;
if(code==1)
printf("\n yellow alert added.");
else if (code==2)
printf("\n orange alert added.");
else
2:2

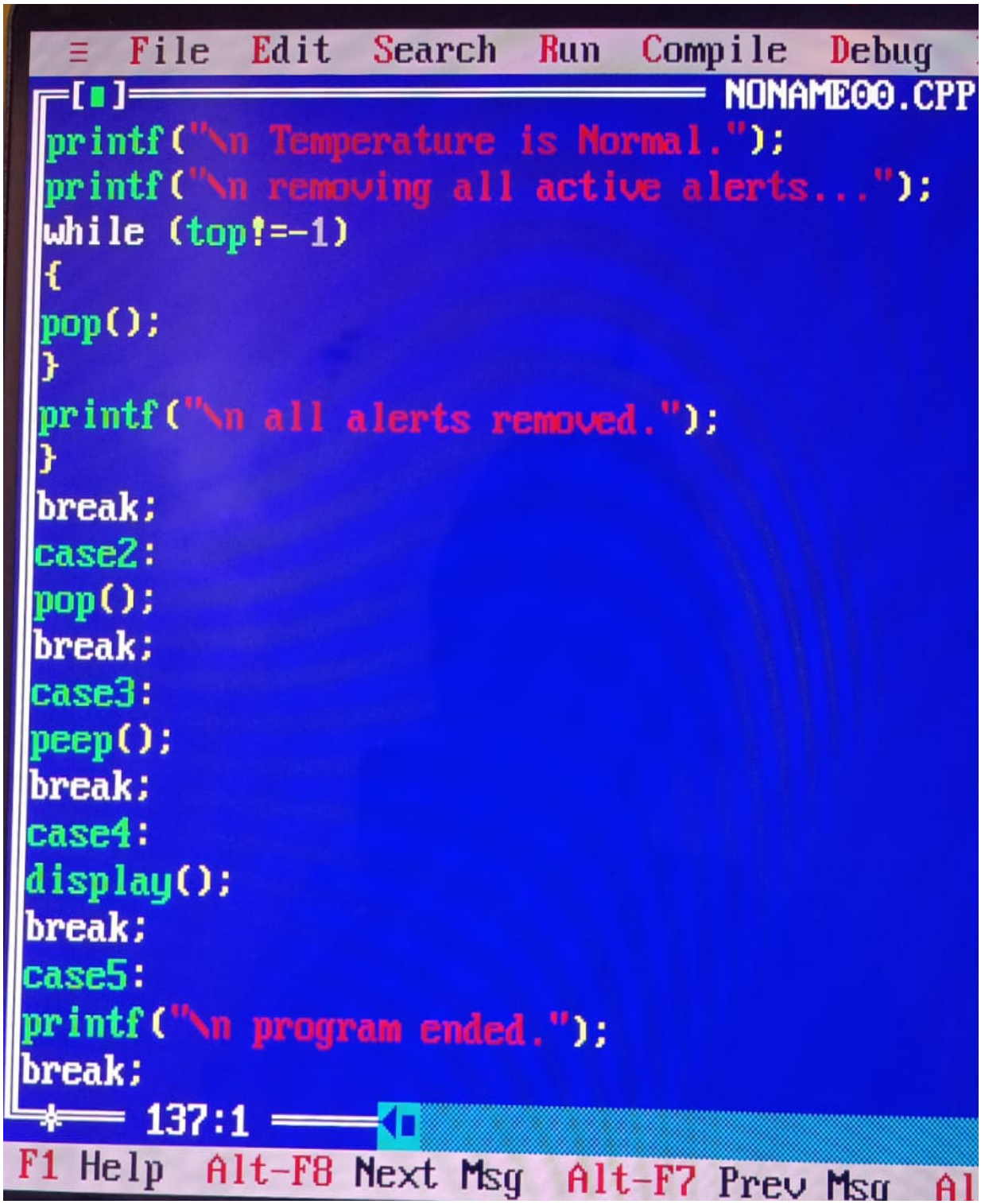

File Edit Search Run Compile
NON
printf("\n orange alert added.");
else
printf("\n red alert added.");
}
}
void pop()
{
if (top==-1)
{
printf("\n stack is empty,");
}
else
{
if(alerts[top]==1)
printf("\n yellow alert removed,");
else if (alerts[top]==2)
printf("\n orange alert removed,");
else
printf("\n red alert removed.");
top--;
}
40:2

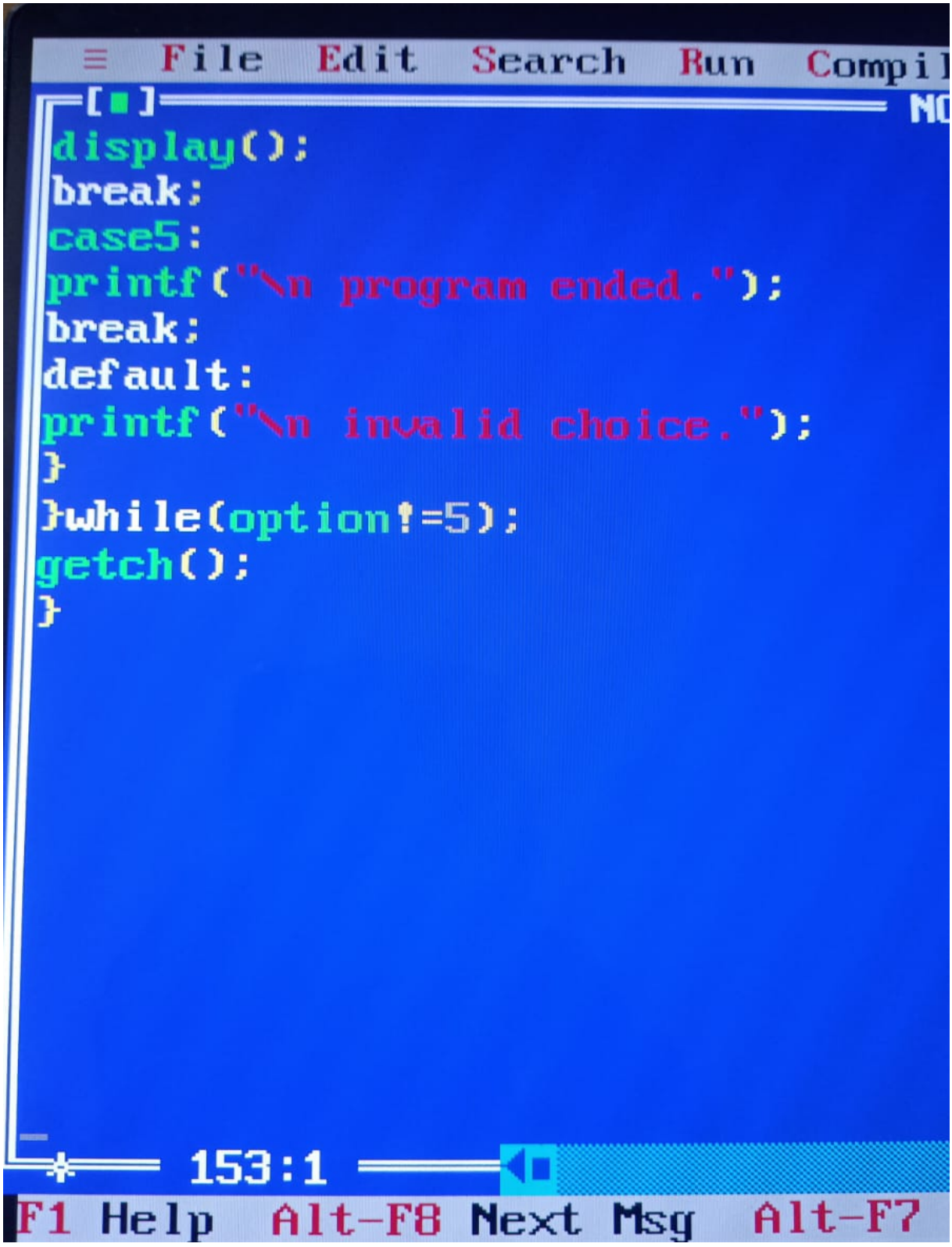

File Edit Search Run Comp
top--;
}
}
void peep()
{
if(top==-1)
{
printf("\n no active alert.");
}
else{
printf("\n top alert:");
if (alerts[top]==1)
printf("yellow alerts");
else if (alerts[top]==2)
printf("orange alert");
else
printf("red alert");
}
}
void display()
{
59:1
F1 Help Alt-F8


File Edit Search Run Compile Debug
NONAME00.C
void display()
{
int j;
if(top==-1)
{
printf("\n no active heatwave alerts,");
}
else
{
printf("\n active heatwave alerts:\n");
for(j=0;j<=top;j++)
{
if(alerts[j]==1)
printf("yellow alert\n");
else if(alerts[j]==2)
printf("orange alert\n"
);
else
printf("red alert\n");
}
}
78:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg


File Edit Search Run Compile Debug
NONAME00.CP
}
void main()
{
int option;
int temp;
clrscr();
do
{
printf("\n\n---HEATWAVE ALERT SYSTEM---");
printf("\n1.Enter Temperature");
printf("\n2.pop alert");
printf("\n3.peep alert");
printf("\n4,Display alerts");
printf("\n5.Exit");
printf("\n\nEnter your choice:");
scanf("%d",&option);
switch(option)
{
case1:
printf("\n Enter Temperature:");
scanf("%d",&temp);
99:1


File Edit Search Run Compile Debug
NONAME00.CP
case1:
printf("\n Enter Temperature:");
scanf("%d",&temp);
if(temp>=45)
{
printf("\n red alert");
push(3);
}
else if (temp>=43)
{
printf("\n orange alert");
push(2);
}
else if (temp>=40)
{
printf("\n yellow alert");
push(1);
}
else
{
printf("\n Temperature is Normal.");
117:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg


File Edit Search Run Compile Debug
NONAME00.CPP
printf("\n Temperature is Normal.");
printf("\n removing all active alerts...");
while (top!=-1)
{
pop();
}
printf("\n all alerts removed.");
}
break;
case2:
pop();
break;
case3:
peep();
break;
case4:
display();
break;
case5:
printf("\n program ended.");
break;
137:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg


File Edit Search Run Compil
display();
break;
case5:
printf("\n program ended.");
break;
default:
printf("\n invalid choice.");
}
}while(option!=5);
getch();
}
153:1
F1 Help Alt-F8 Next Msg Alt-F7

OUTPUT:

```
          --- HEATWAVE ALERT SYSTEM ---

1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit

Enter your choice: 1

Enter Temperature: 41

Yellow Alert
Yellow Alert Added.


          --- HEATWAVE ALERT SYSTEM ---

1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit

Enter your choice: 1

Enter Temperature: 44

Orange Alert
Orange Alert Added.
```

```
         --- HEATWAVE ALERT SYSTEM ---

1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit

Enter your choice: 1

Enter Temperature: 46

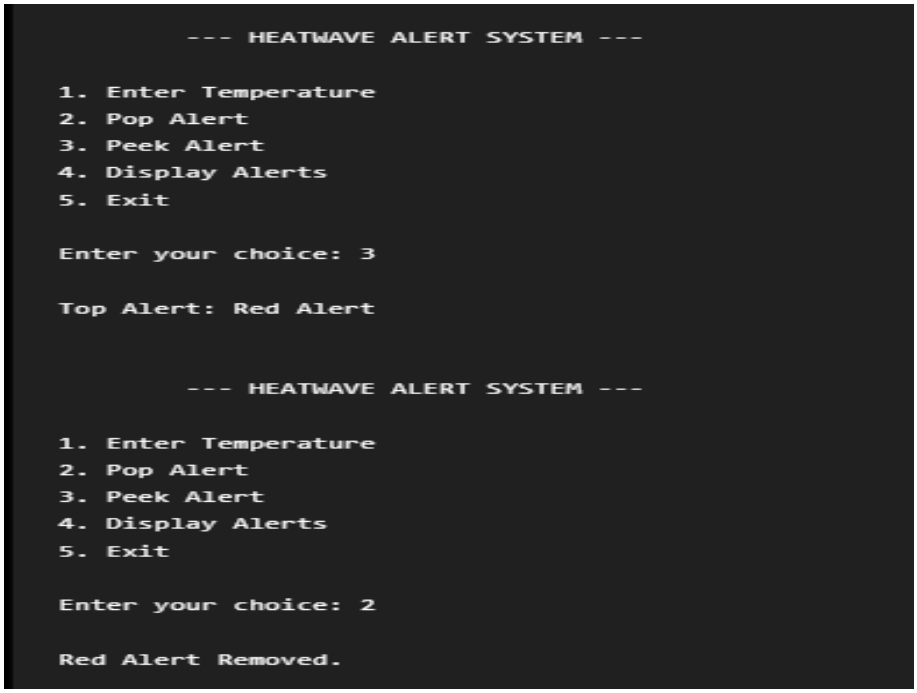
Red Alert
Red Alert Added.

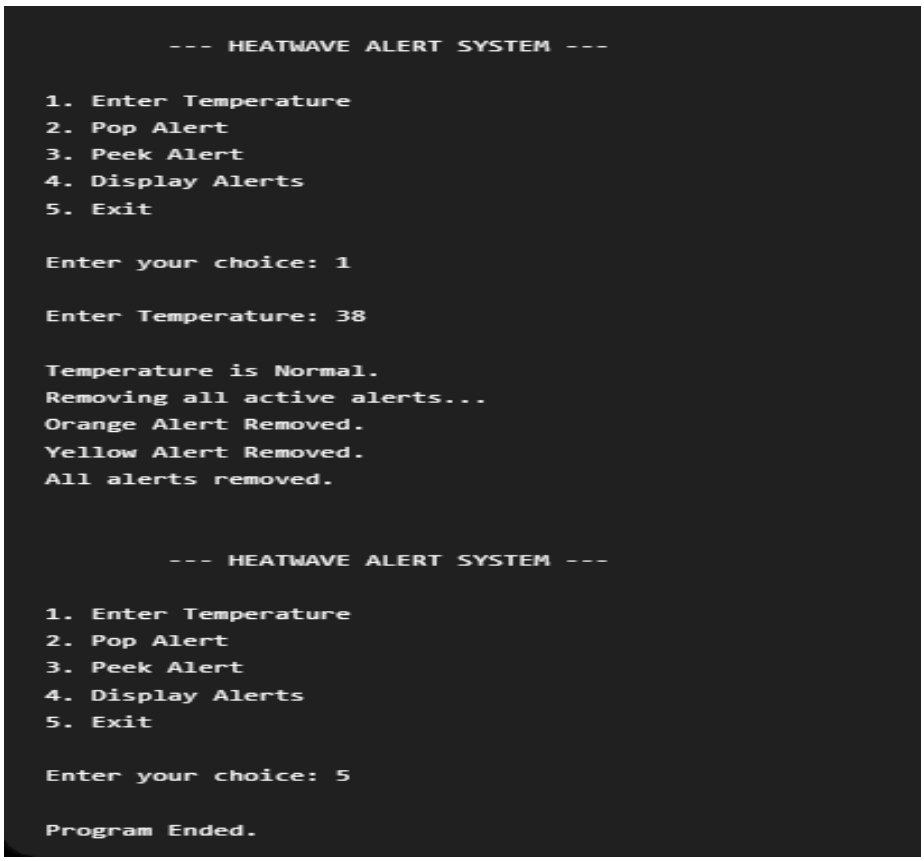

         --- HEATWAVE ALERT SYSTEM ---

1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit

Enter your choice: 4

Active Heatwave Alerts:
Yellow Alert
Orange Alert
Red Alert
```


--- HEATWAVE ALERT SYSTEM ---
1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit
Enter your choice: 3
Top Alert: Red Alert
--- HEATWAVE ALERT SYSTEM ---
1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit
Enter your choice: 2
Red Alert Removed.


--- HEATWAVE ALERT SYSTEM ---
1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit
Enter your choice: 1
Enter Temperature: 38
Temperature is Normal.
Removing all active alerts...
Orange Alert Removed.
Yellow Alert Removed.
All alerts removed.
--- HEATWAVE ALERT SYSTEM ---
1. Enter Temperature
2. Pop Alert
3. Peek Alert
4. Display Alerts
5. Exit
Enter your choice: 5
Program Ended.